SafeRoute Nepal

Proposed Title

SafeRoute Nepal: A Disaster-Aware Road Condition and Route Risk Information System

Project Type

MERN Stack web application + GIS/maps + crowdsourcing + government disaster-data integration.

Problem

During landslides, floods, heavy rainfall, road blockages, and similar events in Nepal, travelers often have difficulty knowing whether a particular road or route is currently usable.

The project combines two kinds of information:

- **Official disaster information** from Nepal's BIPAD system.
- **Local road-condition reports** submitted by users.

Instead of being just another navigation application, SafeRoute Nepal answers:

> **"What hazards or recently reported road problems affect the route I am planning to travel?"**

Nepal's BIPAD platform provides an official API that makes it a particularly useful data source for the project.

BIPAD API: `https://bipadportal.gov.np/api/`

Main Users

SafeRoute Nepal can be used by:

- Travelers
- Local residents
- Delivery drivers
- Public-transport drivers
- Emergency responders
- NGOs
- Local government or ward personnel

Core Features

1. Interactive Nepal Map

An interactive map displays disaster events and road-condition reports across Nepal.

2. BIPAD Integration

SafeRoute Nepal integrates with Nepal's BIPAD system to obtain authoritative disaster information.

3. Crowdsourced Road Reporting

Users can report road-related problems such as:

- Landslides
- Flooding
- Blocked roads
- Damaged roads
- Bridge problems
- Accidents or obstructions
- Construction
- Other hazards

4. Location, Photo, and Timestamp

Each road report can contain a geographic location, supporting photo, and timestamp.

5. Community Verification

Other users can confirm or dispute a submitted road-condition report.

6. Report Expiry

Old reports can automatically expire so outdated road-condition information does not remain active indefinitely.

7. Route Risk Checker

Users enter an origin and destination, and SafeRoute Nepal identifies hazards and road-condition reports located near the selected route.

8. Risk Explanation

SafeRoute Nepal can classify route risk using levels such as:

- **Low**

- **Moderate**
- **High**
- **Severe**

The system should also explain why a particular risk level was assigned.

9. Moderator Dashboard

Moderators can review suspicious, inaccurate, duplicate, or inappropriate reports.

10. PWA and Offline Reporting

Users can create reports when internet connectivity is unavailable. The reports can be stored locally and uploaded when connectivity becomes available again.

Example Scenario

Suppose someone wants to travel:

Kathmandu → Pokhara

SafeRoute Nepal obtains the route and checks nearby:

- BIPAD disaster events
- Landslide locations
- Flood information
- Recent community road reports
- Blocked-road reports
- Report timestamps
- Verification status

The system could produce a result such as:

> **Route Risk: HIGH**
> 2 official hazard events detected
> 3 recent community road reports
> Landslide reported 1.2 km from route
> Road blockage verified by 4 users
> Last updated: 18 minutes ago

The application should describe this information as **risk or advisory information** rather than declaring that a route is completely "safe."

MERN Architecture

Frontend

The frontend can use:

- React
- TypeScript
- MapLibre GL JS
- TanStack Query
- React Hook Form
- Zod
- Turf.js

Turf.js can optionally be used for client-side geospatial calculations.

Backend

The backend can use:

- Node.js
- Express.js
- TypeScript
- REST API
- JWT authentication and authorization
- Scheduled BIPAD synchronization

Database

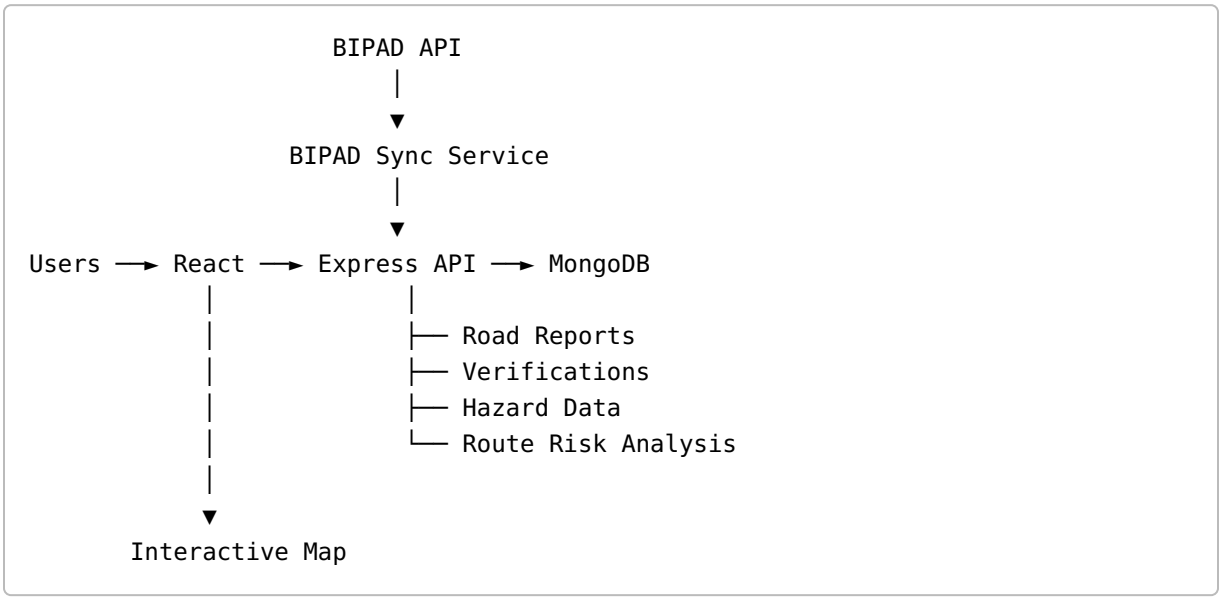
The database can use:

- MongoDB
- Mongoose
- GeoJSON
- `2dsphere` indexes

MongoDB's geospatial functionality allows SafeRoute Nepal to perform queries such as:

- Find reports within a particular distance.
- Find hazards near a route.
- Find reports inside a geographic area.
- Find nearby disaster events.

Basic Data Flow



Main MongoDB Collections

```
users
roadReports
verifications
bipadEvents
routeChecks
alertSubscriptions
moderationActions
syncRuns
```

A road report could conceptually contain:

```
RoadReport
 ├── reporter
 ├── category
 ├── severity
 ├── description
 ├── location (GeoJSON)
 ├── photo
 ├── reportedAt
 ├── expiresAt
 ├── verificationCount
```

```
  ├── confidence
  └── status
```

Recommended Report Categories

SafeRoute Nepal can support the following road-condition categories:

```
Landslide
Flooded Road
Road Blocked
Road Damage
Bridge Problem
Accident / Obstruction
Construction
Other Hazard
```

Route Risk Checker

The **Route Risk Checker** is one of the most important features of SafeRoute Nepal.

This feature makes the project substantially more interesting than a simple disaster map.

The basic process is:

```
Origin
   ↓
Destination
   ↓
Routing Service
   ↓
Route / Polyline
   ↓
Check Nearby BIPAD Hazards
   +
Check Community Road Reports
   ↓
Calculate Risk
   ↓
LOW / MODERATE / HIGH / SEVERE
```

```
   ↓
Explain WHY
```

For an initial implementation, SafeRoute Nepal can use an explainable rule-based algorithm rather than AI.

A rule-based approach is easier to implement, test, evaluate, and demonstrate.

For example:

```
Risk Score =
    Hazard Severity
  × Distance Factor
  × Freshness Factor
  × Confidence Factor
```

The system can then explain why a particular route received a particular risk level.

User Roles

Guest

A guest can:

- View the map.
- View hazards.
- View verified reports.
- Check route risks.

Registered User

A registered user can:

- View the map.
- View hazards and reports.
- Check routes.
- Submit road-condition reports.
- Verify reports submitted by other users.
- Save routes or areas.

Trusted Reporter

A trusted reporter can have greater confidence weighting after submitting a sufficient number of accurate and verified reports.

Moderator

A moderator can:

- Approve reports.
- Reject reports.
- Flag reports.
- Review evidence.
- Handle suspicious reports.

Admin

An administrator can:

- Manage users.
- Manage moderators.
- Inspect BIPAD synchronization.
- View analytics.
- Manage report categories.
- Manage system configuration.

Minimum Viable Product (MVP)

The SafeRoute Nepal MVP should contain:

- User authentication
- Interactive map
- BIPAD disaster data
- Road-condition reporting
- Geographic location support
- Photo support
- Community verification
- Moderator dashboard
- Basic route-risk checking

Advanced Features

After the MVP is working, SafeRoute Nepal can be extended with:

- Offline PWA reporting
- Push notifications
- Trusted-reporter reputation

- Automatic duplicate-report detection
- Advanced analytics
- Nepali and English language support
- Route alternatives
- Optional machine-learning features

The scope should remain focused on **disaster-aware road-condition and route-risk information** rather than attempting to reproduce a complete navigation platform such as Google Maps.

Why SafeRoute Nepal Is a Strong Final-Year Project

SafeRoute Nepal demonstrates more than ordinary MERN CRUD operations.

The project can demonstrate knowledge and implementation of:

- MERN Stack
- REST APIs
- External government API integration
- GIS
- GeoJSON
- Geospatial database queries
- Interactive maps
- Authentication
- Role-Based Access Control (RBAC)
- Crowdsourcing
- Community verification
- Moderation
- Data freshness and expiry
- Background data synchronization
- PWA and offline functionality
- Analytics
- Route analysis
- Risk calculation

SafeRoute Nepal also addresses a clear Nepal-specific real-world problem rather than being a generic management system.

One-Sentence Project Description

> **SafeRoute Nepal is a MERN-based disaster-aware road information and route-risk platform that integrates official disaster data with geotagged community road-**

condition reports to help users identify hazards and disruptions affecting their planned routes in Nepal.

Suggested Final Project Title

SafeRoute Nepal: A Crowdsourced Disaster-Aware Road Condition and Route Risk Information System Using MERN Stack and GIS